

WEEK 3: FUNCTIONAL ANATOMY & KINESIOLOGY

HEAD, NECK & TMJ SKULL

1.) Cranial bones — 8 bones

- Protects brain and internal structure
- Also known as calvarium

LAYERS OF SKULL BONES

- 1.) **External table** - compact bone
- 2.) **Diploe** - trabecular or spongy bone
- 3.) **Internal table** - compact bone
 - **S** - kin
 - **C** - onnective tissue
 - **A** - poneurosis
 - **L** - oose areolar (contains emissary vein; bulbles; most dangerous part)
 - **P** - ericranium

2.) Facial bones — 14 bones

- Forms the structure of face
- Vomer and mandible are unpaired
- Strongest is mandible; weakest is lacrimal (black eye; raccoon eyes)
- Malar bone - zygomatic bone

POINTS

- 1.) **Inin** - philtrum
- 2.) **Bunio** - external occipital protuberance; occiput; bump of knowledge
- 3.) **Pterion** - thinnest part of the skull; contains middle meningeal artery which causes epidural hemorrhage if ruptured
- 4.) **Glabella** - connection between cranial and facial bone; above the skin in between eyebrows

JOINT OF THE SKULL

- Fibrous synarthrodial sutures

CONNECTIVE TISSUE

- **Suture ligament** - connects the bones together

2.) **Posterior occipital** — end of the first year; 12 months

- Posterolateral (mastoid)
 - Late closure: hydrocephalus and mental retardation

FUNCTION

- Allow the growth of the brain; allows the fetal skin to change in size
- Landmark for withdrawal of blood

MUSCLES OF THE FACE

- 1.) **Occipitofrontalis** - moves scalp and raises eyebrow
- 2.) **Orbicularis oculi** - closing the eyes and protecting eyeballs
- 3.) **Corrugator supercilli** - frowning
- 4.) **Compressor nasi**
- 5.) **Dilator naris** - widens nasal aperture
- 6.) **Procerus** - wrinkles the nose bridge
- 7.) **Orbicularis oris** - compress lips together
- 9.) **Levator labii superioris, zygomaticus minor** - elevates the upper lip
- 10.) **Zygomaticus major** - smiling muscle
- 11.) **Levator anguli oris** - draws the corners of the mouth superolaterally
- 12.) **Risorius** - draws the corners of the mouth laterally
- 13.) **Mentalis** - pouting
- 14.) **Buccinator** - compresses cheeks, prevent mouth pocketing
- 15.) **Platysma** - most superficial muscle of neck
- 16.) **TIME** — muscles of mastication
 - **T** - emporalis
 - **I** - nternal pterygoid (medial)
 - **M** - asserter
 - **E** - xternal pterygoid (lateral)
 - TIM closes the mouth
- 17.) **SCM**
 - Unilateral - ipsilateral lateral flexion, contralateral rotation
 - Bilateral - neck flexion, head extension
- 18.) **Suprahyoid muscles**

NEONATAL SKULL

FONTANELLES

1.) **Anterior fontanelle** — closure in 18th month

- Anterolateral sphenoid

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- **G** - geniohyoid
- **S** - stylohyoid
- **M** - mylohyoid
- **D** - digastric (anterior and posterior)

19.) Infrahyoid muscle

- **T** - thyrohyoid
- **O** - omohyoid (superior and inferior)
- **S** - sternohyoid
- **S** - sternothyroid

20.) Scalenes

- Unilateral - ipsilateral lateral flexion and contralateral rotation
- Bilateral - neck flexion

CRANIAL NERVES

1.) Olfactory — sensory

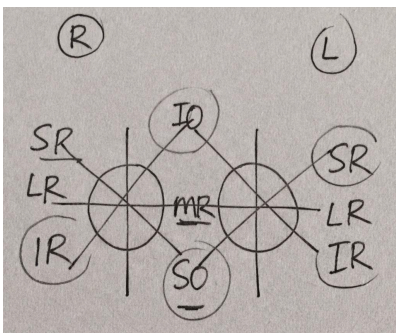
- Sense of smell

2.) Optic — sensory

- Sense of sight
- Afferent for pupillary light reflex

3.) Oculomotor — motor

- Efferent; levator palpebrae superioris (eye opening); constricts pupil
- Extraocular muscles:



- **Oblique muscles** - performs opposite movement relative to its name

4.) Trochlear — motor

- SO4; moves the eyes inferiorly

5.) Trigeminal — mixed

- Mastication; facial sensation of mouth, lips, nose, and tongue; corneal reflex (sensory)

6.) Abducens — motor

- LR6; moves the eyes laterally

7.) Facial — mixed

- Anterior $\frac{2}{3}$ of taste (sweet, salty, sour)
- Muscles of face, cheek, scalp. stylohyoid, and posterior belly of digastric
- Corneal reflex (motor; blinking)
- Ear sensation:

Other name	Muscle
H - hammer	M - malleus
A - anvil	I - incus
S - stirrup	S - tensor

8.) Vestibulocochlear — sensory

- Balance and hearing

9.) Glossopharyngeal — mixed

- Swallowing; stylopharyngeus muscle
- Posterior $\frac{1}{3}$ of taste (bitter)

10.) Vagus — mixed

- Involuntary control of ANS
- Phonation (phonate) - talking

11.) Accessory — motor

1.) Trapezius

- **Upper fiber** - elevates and upwardly rotates scapula
- **Middle fiber** - scapular retraction
- **Lower fiber** - depresses and upwardly rotates scapula

2.) SCM

- neck flexion, head extension, one muscle rotates the head to the opposite side

12.) Hypoglossal — motor

- Tongue movements:

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- 1.) **Styloglossus** - retract or curl
- 2.) **Palatoglossus** - upward movement
- 3.) **Genioglossus** - protrudes
- 4.) **Hyoglossus** - downward

EXITS OF CRANIAL NERVES

- 1.) Cribriform plate of ethmoid - 1
- 2.) Optic canal - 2
- 3.) Superior orbital fissure - 3, 4, 5₁ (first branch of trigeminal nerve)
- 4.) Foramen rotundum - 5₂
- 5.) Foramen ovale - 5₃
- 6.) Internal auditory meatus - 6, 7, 8
- 7.) Jugular foramen - 9, 10, 11
- 8.) Hypoglossal canal - 12

TEMPOROMANDIBULAR JOINT — TMJ

- Location: between the condyle of the mandible and mandibular fossa of the temporal bone
- Normal mastication per day: 1,200 - 500
- Condylod joint:

MOVEMENTS

- 1.) **Protrusion** - less than 7 mm (3-6 mm)
- 2.) **Retrusion** - 3-4 mm
- 3.) **Depression** - 35-55 mm
- 4.) **Elevation**
- 5.) **Lateral deviation or excursion** - 10-15 mm

ARTHROKINEMATIC OF MOUTH OPENING

1.) Early phase

- 35-50% - elevation and depression
 - Predominant movement (rolling & rotation); rolls posteriorly & glides anteriorly

2.) Late phase

- 50-65%
 - Anterior and inferior translation

FUNCTIONAL NORMAL MOUTH OPENING

- 40 mm = 25 mm of rotation + 15 mm translator glide

JAW DEVIATION

1.) Right jaw deviation

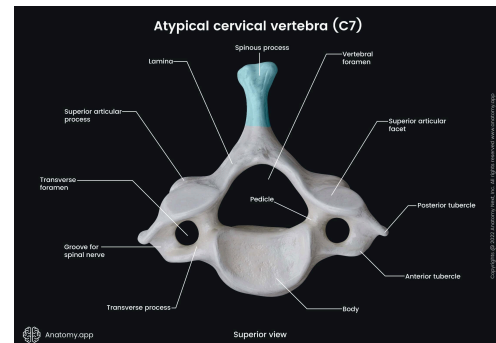
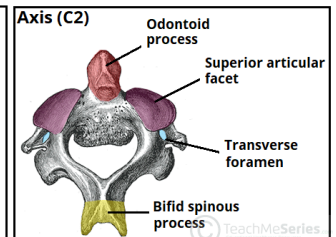
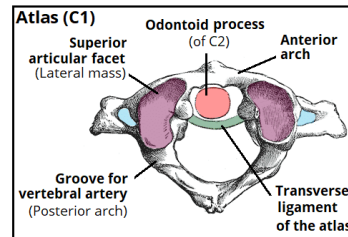
- Contract: L lateral pterygoid, R medial pterygoid
- Relax: R lateral pterygoid, L medial pterygoid

NECK

CERVICAL SPINE

1.) Atypical spine

- C1: no body; no spinous process
- C2: has dens and odontoid process
- C7: has vertebral prominence; long non-bifid spinous process



2.) Body of cervical spine

- Small and broad
 - Vertebral foramen: triangular due to the movement of esophagus
 - Facet
 - Transverse process
 - Between spine:
 - Inferior facet - oriented superiorly, medially, and anteriorly

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- Superior facet - oriented inferiorly, laterally, and posteriorly

3.) Movement of cervical spine

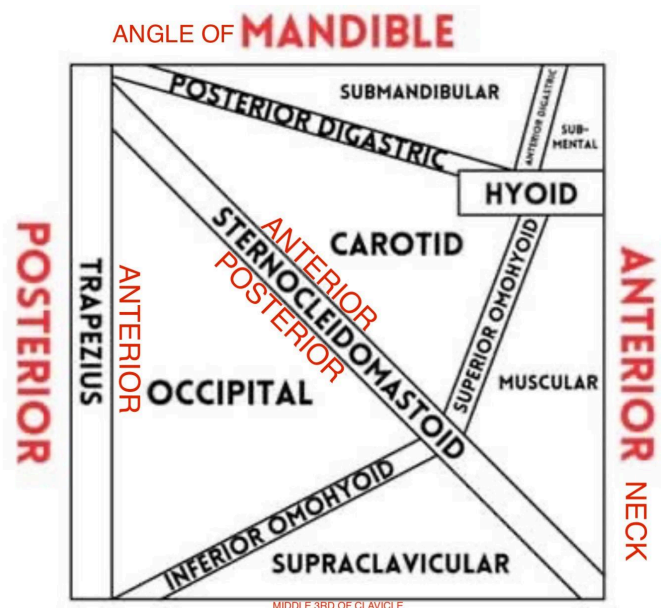
- Zygapophyseal joints

- Facets are oriented 45° to the transverse/frontal plane

Movement	C0-C1	C1-C2	C2-C7
Extension	Roll Posteriorly, Glide Anterior	C1 tilts posteriorly on C2	Body of C3 tilts posteriorly on C4; IAF of C3 slides posterior and inferiorly on SAF of C4
Flexion	Roll Anterior, Glide Posterior	C1 tilts anterior on C2	Body of C3 tilts anteriorly on C4; IAF of C3 slides anterior and superior on SAF of C4
Lateral	Roll Right, Glide Left	Negligible	
Rotation	Negligible		Body of C3 rotates on C4. Rotation right: (L) IAF of C3 slides superior/anteriorly on SAF of C4; (R) IAF of C3 slides inferiorly/posteriorly on SAF of C4

LIGAMENTS

KIND	FUNCTION
1.) TMJ ligament	Strongest; limits posterior and downward movement of mandible
2.) Stylomandibular	Weakest; connects styloid process and mandible; limits protrusion
3.) Sphenomandibular	Connects sphenoid and mandible; limits lateral deviation and translation
4.) Anterior atlantooccipital membrane	
5.) Posterior atlantooccipital membrane	
6.) Apical	Connect C1 and skull
7.) Alar	Detach dens; limits rotation; limits AA rotation
8.) Cruciate	Limits anterior and posterior rotation; limits AA rotation
9.) Membrane tectoria	Limits flexion
10.) ALL	Connects anterior part of spine; limit extension
11.) PLL	Connects posterior part of spine; limit flexion
12.) Ligamentum nuchae	Nuchal line and occiput; limits flexion
13.) Intertransverse ligament	Connects transverse processes; limits lateral rotation and flexion
14.) Ligamentum flavum	Yellow in color due to fat



1.) Anterior triangle

- **Superior:** angle of mandible; **anterior:** anterior neck; posterior SCM

2.) Posterior triangle

- **Anterior:** posterior SCM; **posterior:** posterior trapezius; **Inferior:** middle 3rd of clavicle

TRIANGLE OF NECK